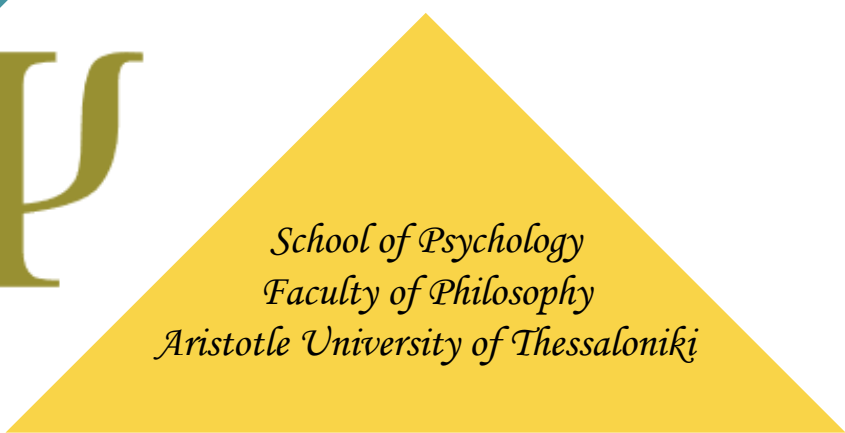




Exploratory Factor Analysis (EFA)

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Exploratory Factor Analysis

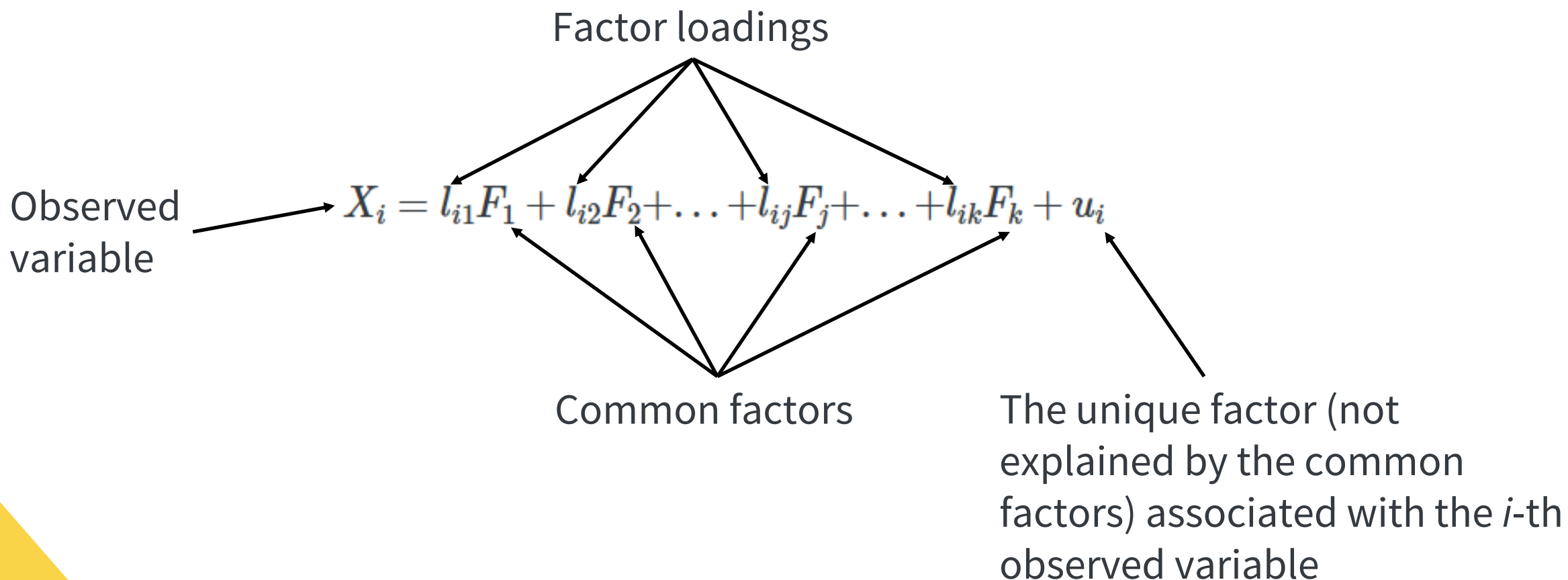
Exploratory Factor Analysis (EFA) focuses on identifying unobserved **latent variables** (also known as common factors) that explain the correlations among observed variables, with the aim of uncovering the data's underlying structure.

- **Observed variables** (also called manifest variables) are the variables that we can directly measure or observe.
- **Latent variables** are hidden or abstract concepts inferred from manifest variables.

Exploratory Factor Analysis (EFA) can be used to explore the dimensionality of a measurement instrument (e.g., questionnaire) by identifying the smallest number of interpretable factors that explain the **common variance** among the observed variables.

Exploratory Factor Analysis model

The factor model for the i -th observed variable (X_i) for a single individual can be expressed as a **regression on the common factors**:



Steps in the process of EFA

1. Prepare the Data (standardization, outliers)
 2. Evaluate Assumptions (Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity, multicollinearity or singularity)
 3. Choose the Extraction Method (e.g., Principal Axis Factoring, Maximum Likelihood, or Maximum Residuals).
 4. Determine the Number of Factors (eigenvalues greater than one, scree plot inspection, or parallel analysis).
 5. Factor Rotation (Orthogonal Vs Oblique)
 6. Interpret the Factors
 7. Assess Factor Reliability (Cronbach's Alpha >0.7)
 8. Refine the Model
 9. Finalize the Factor Solution
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Example of GRMS Stress Appraisal

25 Items

- Unattractive because of size of butt (Obj1)
- Negative comments about size of facial features (Obj2)
- Imitated the way they think Black women speak (Obj3)
- Someone made me feel unattractive (Obj4)
- Negative comment about skin tone (Obj5)
- Someone assumed I speak a certain way (Obj6)
- Objectified me based on physical features (Obj7)
- Someone assumed I have a certain body type (Obj8)
- Made a sexually inappropriate comment (Obj9)
- Negative comments about my hair when natural (Obj10)
- I have felt unheard (Marg1)
- My comments have been ignored (Marg2)
- ...

EFA Assumptions

Identity matrix

$$I_5 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Bartlett's test of sphericity**

A significant result ($p < 0.05$) indicates that the correlation matrix significantly differs from an identity matrix. This suggests that the variables share enough correlation to justify the use of EFA).

- **No multicollinearity and singularity**

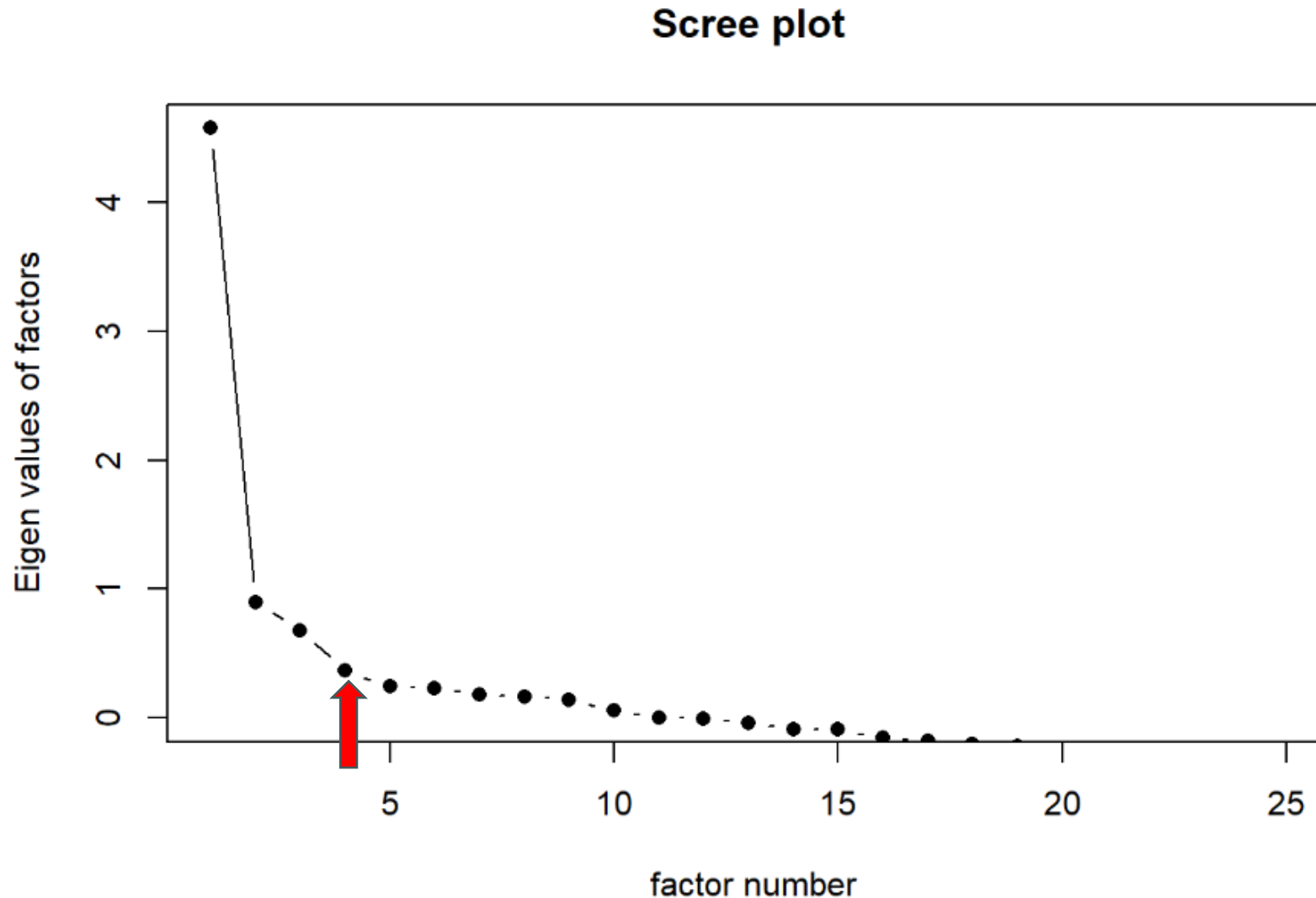
- **Kaiser-Meyer-Olkin (KMO) index of Sampling Adequacy**

(bare minimum of 0.5, values between 0.5 and 0.7 as mediocre, values between 0.7 and 0.8 as good, values between 0.8 and 0.9 as great, values above 0.9 are superb)

- **Address outliers.**

Scree Plot-Number of factors

Determine the number of factors to retain in EFA. We will select the first four factors.



Factor loadings

Common factors
Standardized loadings (pattern matrix) based upon correlation matrix

	PA1	PA2	PA3	PA4	h2	u2	com
Obj1	0.508				0.293	0.707	1.06
Obj2	0.526				0.319	0.681	1.04
Obj3	0.463				0.250	0.750	1.07
Obj4	0.563				0.326	0.674	1.04
Obj5	0.473				0.280	0.720	1.17
Obj6	0.385			0.246	0.225	0.775	1.77
Obj7	0.535				0.273	0.727	1.43
Obj8	0.391				0.353	0.647	1.52
Obj9	0.479				0.186	0.814	1.32
Obj10					0.217	0.783	1.15
Marg1		0.811			0.619	0.381	1.01
Marg2		0.492			0.411	0.589	1.52
Marg3					0.277	0.723	1.19
Marg4					0.281	0.719	1.82
Marg5					0.326	0.674	1.26
Marg6	0.310	0.250			0.271	0.729	2.31
Marg7		0.356			0.181	0.819	1.20
Str1			0.558		0.363	0.637	1.16
Str2			0.266		0.179	0.821	2.51
Str3					0.276	0.724	1.01
Str4					0.167	0.833	1.19
Str5					0.178	0.822	1.40
Ang1			0.334	0.241	0.223	0.777	2.15
Ang2					0.331	0.569	1.03
Ang3				0.298	0.195	0.805	1.91

Loadings

Loadings

Loadings

Loadings

Uniqueness represents the portion of variance that is specific to the variable itself and **not explained** by the common factors.

Communality represents the proportion of an item's total variance accounted for by the common factors.

The SS Loadings column represents the **eigenvalues** λ for each factor (before rotation).

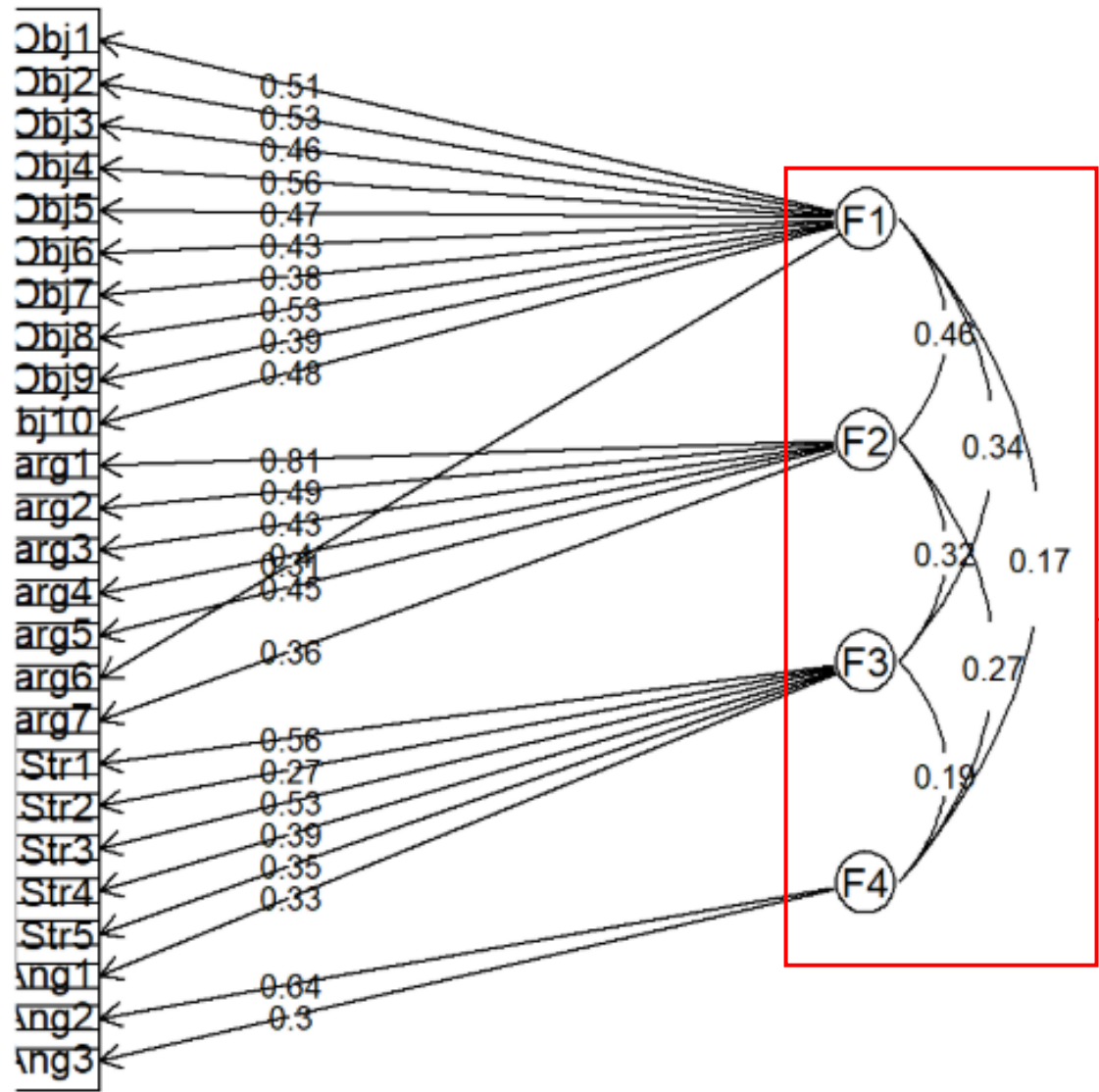
Explained variance

Factor	SS_Loadings	% of Var	Cum Var (%)
1 PA1	2.79	11.2	11.2
2 PA2	2.03	8.1	19.3
3 PA3	1.37	5.5	24.8
4 PA4	0.91	3.7	28.4

A cumulative common variance of 28.4% explained by the first four factors

Factor Analysis diagram-Oblimin rotation

Factor Analysis



- **Factor 1: Assumptions of Beauty and Sexual Objectification** (11 items; Obj1-Obj10 and Marg6)
- **Factor 2: Silenced and Marginalized** (6 items; Marg1-Marg5, Marg7)
- **Factor 3: Strong Woman Stereotype** (6 items; Str1-Str5, Ang1)
- **Factor 4: Angry Woman Stereotype** (2 items; Ang2 and Ang3)

Inter-factor Correlations					
	Factor	PA1	PA2	PA3	PA4
1	PA1	1.00	0.46	0.34	0.17
2	PA2	0.46	1.00	0.32	0.27
3	PA3	0.34	0.32	1.00	0.19
4	PA4	0.17	0.27	0.19	1.00

The four factors are positively correlated with each other, with correlations ranging from 0.17 to 0.46

Model fit

- Chi-square test. If $p > 0.05$, the model **fits the data well** (fail to reject H_0).
- Root mean square error of approximation (**RMSEA**). It is suggested $RMSEA < 0.05$ (but definitely < 0.10) for a good fit.

RMSEA Lower Upper				TLI	BIC	Chi2	df	p
RMSEA	0	0	0.02	1.03	-955.52	189.19	206	0.794



An RMSEA value close to 0, indicating a good fit between the model and the observed data.



Model fits the data well ($p > 0.05$)

Reliability analysis

Cronbach's alpha is a psychometric metric used to measure the internal consistency reliability of a multi-item scale or questionnaire.

It asks a simple question: If a group of questions is supposed to measure the exact same underlying construct (factor), do participants answer them in a consistent manner?

	Factor	Items	Cronbach_alpha
1	F1	11	0.778
2	F2	6	0.731
3	F3	6	0.594
4	F4	2	0.397

Note: A factor with more items can artificially inflate Cronbach's alpha.